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INTERNATIONAL ACADEMIC RESEARCH COMPETITION (IARCO) 2026

RESEARCH PROPOSAL SUBMISSION

TITLE OF THE RESEARCH PROPOSAL	In Silico Generative Design and Molecular Dynamics Validation of Novel Antimicrobial Peptides Targeting Carbapenem-Resistant Acinetobacter baumannii
FULL NAME	Gustavo Naves Ávila de Moura
INSTITUTION	Zênite
COUNTRY	Brazil
SUBMISSION DATE	September 21, 2026,
CATEGORY	Junior

In Silico Generative Design and Molecular Dynamics Validation of Novel Antimicrobial Peptides Targeting Carbapenem-Resistant Acinetobacter baumannii

Abstract

Antimicrobial resistance (AMR) is a severe threat to global healthcare, and the World Health Organization has designated Carbapenem-Resistant Acinetobacter baumannii (CRAB) as a Priority 1 Critical Pathogen. Standard high-throughput drug screening methods are severely hindered by costs in the billions of dollars, durations spanning many decades, and very high rates of human cytotoxicity. To overcome these fundamental obstacles, this research proposal introduces an end-to-end computational pipeline that combines generative deep learning with biophysical simulations in order to design new antimicrobial peptides (AMPs) which have high target specificity and low eukaryotic toxicity. The approach involves adapting Meta's ESM-2 protein language model to generate new amphipathic peptide sequences. The candidate sequences are then passed through a multi-stage filtering procedure using Quantitative Structure-Activity Relationship (QSAR) classifiers so that only those with sub-micromolar bactericidal activity are retained and those with hemolytic or proteolytic problems are discarded. The promising candidates are modelled using ESMFold and then subjected to 100-nanosecond atomistic Molecular Dynamics (MD) simulations in GROMACS, with a realistic asymmetric outer membrane model of A. baumannii being used. The binding free energies (ΔGbind) are obtained through MM/PBSA calculations in order to evaluate both the thermodynamic stability and the mechanism of membrane disruption. Because the method reduces the time required to identify early-stage candidates from years to just weeks, it provides a scalable, open-source strategy for designing therapeutic leads directed at critical multi-drug resistant pathogens.

1. Research Problem

Antimicrobial resistance (AMR) is one of the most serious public health issues of the 21st century. The World Health Organization (WHO) has predicted that if new treatments are not rapidly introduced into clinical practice, drug-resistant complications could result in 10 million deaths every year globally by 2050 [1]. Among the multi-drug resistant (MDR) hospital-acquired pathogens, Acinetobacter baumannii, especially the carbapenem-resistant strains (CRAB), has been designated as a Priority 1: Critical pathogen [2]. CRAB poses a serious threat in intensive care units, causing severe instances of

hospital-acquired pneumonia, septicemia, and wound infections, and patient mortality rates are frequently above 60% since these bacteria are highly resistant to almost all first-line beta lactams, carbapenems, and the last-resort polymyxins [3]. The primary difficulty in dealing with CRAB is the current method of discovering small-molecule antibiotics. It requires 10 to 15 years and capital expenditures of over $1 billion per drug to start from scratch and bring a new chemical antibiotic to clinical use [4]. Furthermore, the development of small-molecule drugs has a very high failure rate more than 90% of them failing during the preclinical and clinical phases mainly due to off-target toxicity towards human cells, poor bioavailability, and the rapid emergence of bacterial mutations at the target site [4]. As traditional antibiotics work on a single enzymatic pathway (for example, cell wall synthesis or ribosome translation), bacteria are able to evolve point mutations that eliminate the effect of these drugs. For this reason, there is an urgent need for computational methods that can be used to design therapeutic agents which cause physical disruption, such as Antimicrobial Peptides (AMPs), in order to avoid the long and tedious trial-and-error processes that involve wet-lab experiments.

2. Existing Literature & Knowledge Gap

Antimicrobial Peptides (AMPs) are naturally occurring, short, positively charged and amphipathic sequences made up of amino acids (with a typical length of 12 to 30 amino acid residues) and form an important component of the innate immune defence in eukaryotic organisms [5]. Unlike conventional small molecules, AMPs destroy bacteria by means of a physical process: the positively charged groups in them bind selectively to the negatively charged lipopolysaccharide (LPS) and phospholipids in the outer membrane of the bacteria, causing pore formation, membrane permeabilisation, and quick leakage inside the cell [6]. Since this mechanism acts on the fundamental biophysical properties of the membrane rather than on specific proteins, it is very difficult for bacteria to evolve resistance to them [6].

Despite their therapeutic potential, the clinical translation of wild-type AMPs has been severely impeded by two main bottlenecks:

- With regard to systemic toxicity and hemolysis: many natural AMPs do not have strict selectivity for cells and cause the lysing of human red blood cells (erythrocytes) at therapeutic doses [7].
- Peptides made up of L-amino acids are not stable in the plasma and are quickly broken down by the proteases present in mammalian serum [8].

Recent advances in deep learning have led to the development of Protein Language Models (pLMs), for example Meta's ESM-2 (Evolutionary Scale Modeling), which acquire rich structural and functional representations from hundreds of millions of unannotated protein sequences [9]. Although generative models are able to suggest new peptide sequences, the current methodologies still have a significant knowledge gap in that they produce candidates solely on the basis of sequence probability without incorporating explicit biophysical toxicity filters or high-resolution dynamic modelling of membrane interactions under physiological conditions. Most of the studies published to date carry out static molecular docking, and this approach is unable to account for the dynamic structural changes, the mechanics of insertion, and the thermodynamic stability of AMPs within realistic bacterial membrane bilayers.

3. Research Question & Objectives

Central Research Question

"Can a generative process guided by deep learning and using fine-tuned transformer language models design new antimicrobial peptides that have a binding affinity of less than one micromolar for the membranes of

carbapenem-resistant Acinetobacter baumannii while still exhibiting almost no hemolytic toxicity when tested in silico?"

Primary Objective

We need to develop, carry out, and assess an integrated computational discovery pipeline which produces new cationic AMPs and checks their thermodynamic stability, cell selectivity, and ability to disrupt membranes of A. baumannii using quantitative machine learning filters and atomistic Molecular Dynamics (MD) simulations.

Specific Objectives

1. For the purpose of dataset curation, construct a database of verified anti-Gram-negative AMPs that is free of redundancy by obtaining the data from public repositories (DBAASP v3.0, APD3, UniProt).
2. Use PyTorch to fine-tune the ESM-2 transformer architecture (esm2_t33_650M_UR50D) in order to produce new, non-natural peptide sequences that have been optimised for a cationic, amphipathic alpha-helical structure.
3. Use machine learning classifiers (ToxinPred3 and Hemolytik) to filter the raw candidate sequences and select those with a low Minimum Inhibitory Concentration (MIC ≤ 4 μg/mL) and a high predicted hemolytic threshold (HC50).
4. To carry out biophysical validation, predict the three-dimensional tertiary structure using ESMFold and then perform 100-nanosecond atomistic Molecular Dynamics (MD) simulations with GROMACS against an asymmetric model of the A. baumannii outer membrane in order to calculate the binding free energy (ΔGMM/PBSA) and determine the insertion mechanics.

4. Research Topic & Innovation

This research proposal lies at the point where Artificial Intelligence, Structural Bioinformatic Engineering, and Infectious Disease Therapeutics come together.

Key Innovations:

- In the case of Generative AI Sequence Exploration, rather than relying on the chemical synthesis of random mutants, the project samples high-probability therapeutic motifs by using fine-tuned protein transformers to browse the infinite combinatorial sequence space (that of 2020 possible 20-mer peptides) within hours.
- The system features a two-level approach to toxicity: it incorporates machine learning QSAR (Quantitative Structure-Activity Relationship) classifiers directly into the loop used for selecting candidates before carrying out structure modeling, and this actively gets rid of any candidates that are cytotoxic at an early stage in order to avoid high preclinical failure rates.
- Realistic asymmetric membrane dynamics: Unlike conventional computational studies in which peptides are docked onto homogeneous, non-physiological lipid bilayers, this protocol involves modelling a complex, asymmetric outer membrane of A. baumannii containing Lipid A, KDO2, and POPE/POPG lipid matrices in a solvated, ionic environment.

5. Methodology

The research method that is being proposed is made up of four sequential and fully reproducible computational stages, using open-source bioinformatics tools.

Phase 1: Data Curation & Preprocessing

- The antimicrobial peptides that are known to act against Gram-negative pathogens will be obtained

from DBAASP v3.0 and APD3.

- The filtering criteria are that the sequences must be cationic peptides with a net charge ranging from +2 to +6, a length between 12 and 30 residues, and a minimum inhibitory concentration of 16 μg/mL or less against A. baumannii.
- Redundancy is reduced using the CD-HIT algorithm with a sequence identity threshold of 90%.

Phase 2: Generative Model Fine-Tuning

- Choice of model: the pre-trained transformer model esm2_t33_650M_UR50D (650M parameters).
- Fine-tuning was carried out using Python and the PyTorch/HuggingFace toolkit in order to adjust the attention weights with respect to cationic and amphipathic motifs.
- Sampling. We used temperature-controlled sampling with T = 0.7 to generate 1,000 new candidates.

Phase 3: Physicochemical & Toxicity Screening (QSAR Filtering)

- The hydrophobic moment (μH ≥ 0.45) and the net charge (+3 to +6) were assessed using Biopython and Heliquest.
- The toxicity classifiers ToxinPred3 and Hemolytik were used in order to eliminate the compounds that are cytotoxic and hemolytic.
- The stability of plasma is predicted by using the PROT-STAB algorithms for determining the half-life of serum.

Phase 4: Structural Modeling, Docking, & Molecular Dynamics Simulations

- The tertiary structure was produced using ESMFold.
- The docking of membranes using AutoDock Vina and HADDOCK 2.4 was performed on the headgroups of Lipid A.
- The molecular dynamics simulation (using GROMACS) involved a complex that had been assembled through CHARMM-GUI in the form of an asymmetric bilayer (composed of KDO2-Lipid A / POPE / POPG). The simulation was carried out using the CHARMM36m force field, TIP3P water, with a salt concentration of 0.15M NaCl at a temperature of 310.15 K and a pressure of 1 bar. The analysis was performed using RMSD, RMSF, data on membrane insertion depth, and the MM/PBSA binding free energy (ΔGbind).

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